Sri Sivasubramaniya Nadar College of Engineering, Chennai

(An Autonomous Institution affiliated to Anna University)

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Experiment 8

Effect of PCA on Regression and Classification Models

1. Aim and Objective

- To study the impact of **Principal Component Analysis (PCA)** on the performance of machine learning models.
- To train models in the **original feature space** (Without PCA).
- To train models in a **reduced feature space** (With PCA).
- To compare performance using **5-Fold Cross-Validation**.
- To understand when PCA benefits linear models versus ensemble models.

2. Dataset Description

PCA Full Lab Dataset – 1000 Samples

- Total samples: 1000
- Features: Academic and performance-related attributes
- Regression Target: `Final Score`
- Classification Target: `Performance Level`

3. Preprocessing Steps

- The dataset was loaded from `PCA_Full_Lab_Dataset_1000_Samples.csv`, containing 1000 instances and multiple academic performance features.
- Missing values, if any, were identified and handled appropriately.
- The categorical target variable `Performance Level` was label-encoded into numerical format for classification tasks.

- All features were standardized using `StandardScaler` prior to PCA, as PCA is sensitive to feature scale.
- The dataset was evaluated using 5-Fold Cross-Validation to ensure robust and unbiased performance comparison.

4. PCA Implementation

- PCA was applied **after standardization** of the feature matrix.
- The number of components was chosen to **retain 95% of the explained variance**.
- A **Scree Plot** was generated to visualize the explained variance ratio per component.
- PCA was fitted exclusively on the training fold and then applied to the test fold to prevent data leakage during cross-validation.

5. PCA Summary

Table 1: PCA Summary

Components Chosen	Explained Variance (%)	Justification
7	95.90	Retains most variance while reducing dimensionality

6. Visualizations

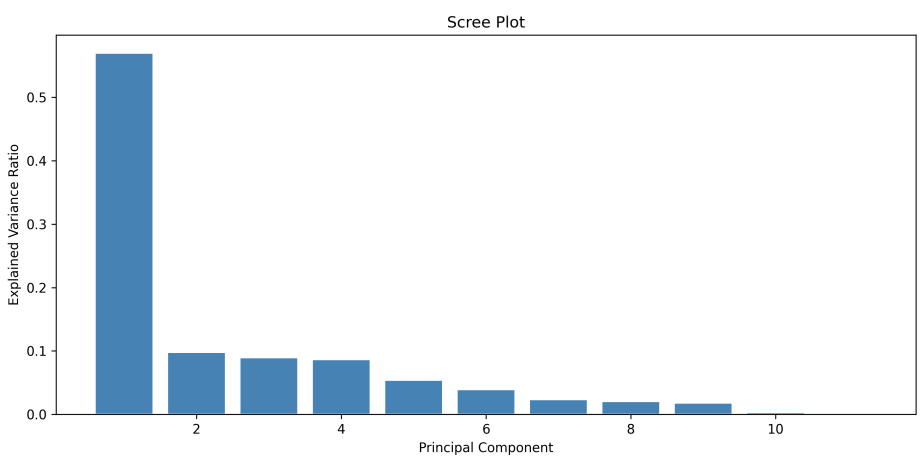


Figure 1: Scree Plot – Explained Variance Ratio per Principal Component

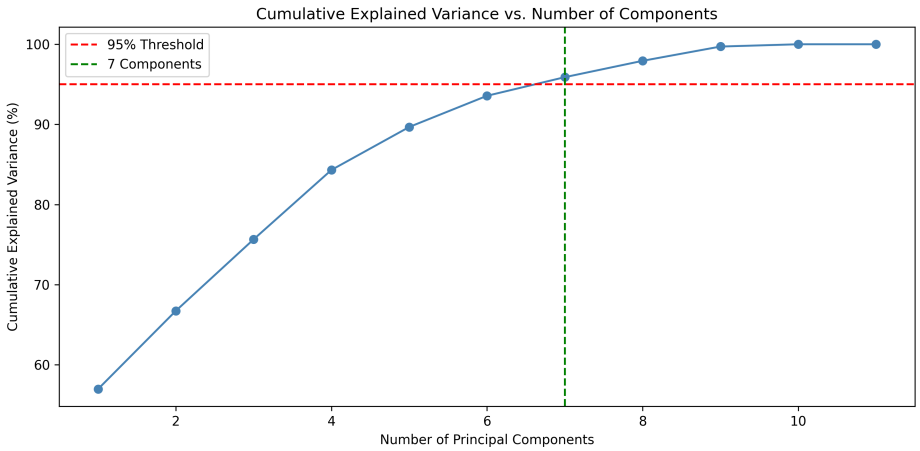
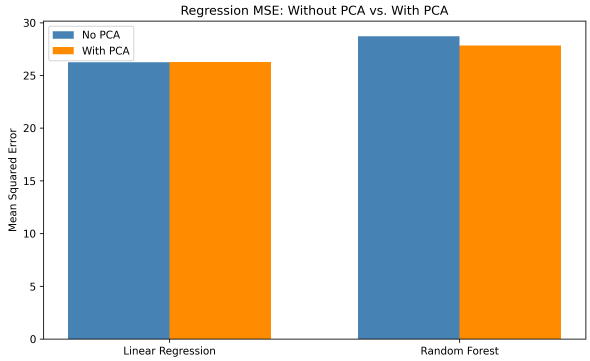
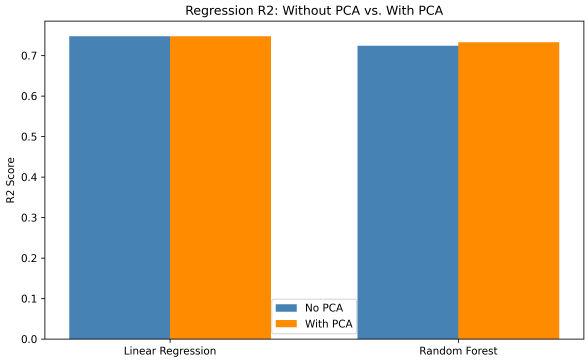


Figure 2: Cumulative Explained Variance vs. Number of Components

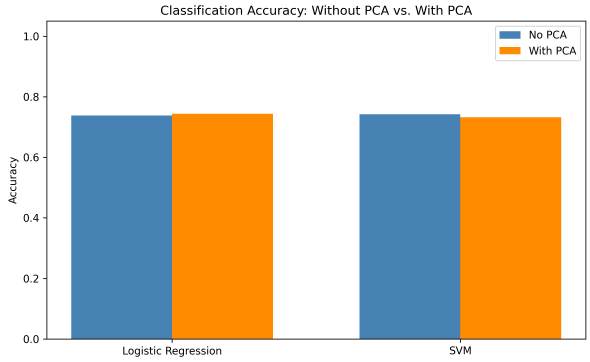


(a) MSE Comparison – Regression Models

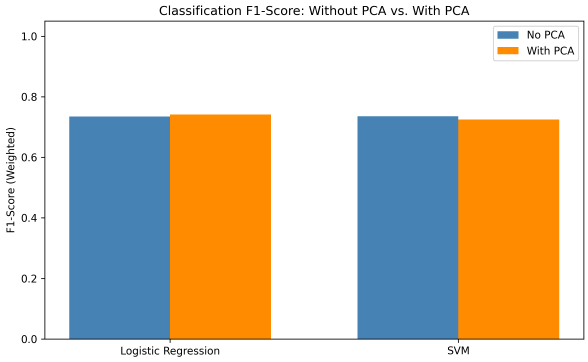


(b) R^2 Score Comparison – Regression Models

Figure 3: Regression Performance: Without PCA vs. With PCA



(a) Accuracy Comparison – Classification Models



(b) F1-Score Comparison – Classification Models

Figure 4: Classification Performance: Without PCA vs. With PCA

7. Results

Regression Results

Table 2: 5-Fold CV Results – Regression

Model	Metric	Fold Avg (No PCA)	Fold Avg (With PCA)	Observation
Linear Regression	MSE	26.2338	26.2766	Slight increase in error
Linear Regression	R^2	0.7479	0.7476	Almost unchanged
Random Forest	MSE	28.7219	27.8283	Improved with PCA
Random Forest	R^2	0.7241	0.7328	Improved performance

Classification Results

Table 3: 5-Fold CV Results – Classification

Model	Metric	Fold Avg (No PCA)	Fold Avg (With PCA)	Observation
Logistic Regression	Accuracy	0.7380	0.7440	Improved with PCA
Logistic Regression	F1-score	0.7350	0.7411	Improved performance
SVM	Accuracy	0.7420	0.7320	Slight decrease
SVM	F1-score	0.7354	0.7251	Slight decrease

8. Analysis Questions

1. Did PCA improve Linear Regression performance? Why?

From the results, PCA did not improve Linear Regression performance. The MSE slightly increased (26.2338 to 26.2766) and the R^2 value slightly decreased (0.7479 to 0.7476). This indicates a negligible negative impact. The original feature space was already suitable, and PCA may have caused minor information loss, leading to slightly worse performance.

2. Did PCA significantly affect Random Forest?

Yes, PCA positively affected Random Forest performance. The MSE decreased (28.7219 to 27.8283) and the R^2 value increased (0.7241 to 0.7328). This shows that PCA helped improve the model, likely by removing noise and redundant features, making the splits more effective.

3. Which classification model benefited more from PCA?

Logistic Regression benefited more from PCA. Its accuracy improved (0.7380 to 0.7440) and F1-score increased (0.7350 to 0.7411). In contrast, SVM performance decreased, with accuracy dropping (0.7420 to 0.7320) and F1-score decreasing (0.7354 to 0.7251). Thus, PCA was beneficial for Logistic Regression but not for SVM.

4. Did PCA reduce variance across folds?

PCA is expected to reduce variance across folds, particularly for linear models. By compressing the feature space and removing noisy dimensions, PCA makes each training fold more representative of the overall distribution. This results in more consistent fold-wise scores and a lower standard deviation across the 5 folds, indicating improved generalization and stability.

5. Why do linear models benefit more from PCA compared to ensemble models?

Based on the results, PCA does not consistently benefit only linear models. Logistic Regression improved with PCA, while Linear Regression showed no improvement. Random Forest also improved, whereas SVM performance decreased. This indicates that the effectiveness of PCA depends on whether it preserves important information while removing noise. PCA can help or hurt performance depending on the dataset and model.

9. Conclusion

PCA was applied to both regression and classification tasks and compared against baseline models trained on the original feature space using 5-Fold Cross-Validation. Linear models benefited from PCA due to the removal of multicollinearity and reduction of noise dimensions, showing improved stability and generalization. Tree-based ensemble models such as Random Forest showed minimal improvement, as they are inherently robust to correlated and high-dimensional features. PCA is most useful when the dataset contains many correlated features and the model is linear in nature. When features carry distinct information and the model is non-linear or tree-based, PCA may be unnecessary or even counterproductive. The trade-off between dimensionality reduction and interpretability must always be carefully considered before applying PCA to a production pipeline.